# The Polyphonic Model

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# Creative chat conversations

- Creativity fostering → Brainstorming using chat
- Micro-creativity → Computer Supported Collaborative Learning

# Discourse

- Monologue
  - Unidirectional model of communication, from a speaker to a listener – theories and methods of analysis:
    - Rhetorical Schema Theory
    - Centering Theory
    - Co-reference resolution
- Dialogue
  - Phone-like, face to face style – units of analysis:
    - Speech acts
    - Dialog acts
    - Adjacency pairs
- Multi-parties conversation

# Analysis methods in literature

- TF-IDF
- Latent Semantic Analysis
- Latent Dirichlet Allocation
- Naïve Bayes
- Social Network Analysis
- Semantic distances using WordNet
- Support Vector Machines
- Collin's perceptron
- TagHelper environment
- (Deep) Neural Networks

# Language, music, and neurology

# Music and language

- “Where the word fails, remains the music [...] Ideas are dead melodies” (Cioran, 2016)
- “If a composer could say what he had to say in words he would not bother trying to say it in music.” (Gustav Mahler)
- “Music is the universal language of mankind.” (Henry Wadsworth Longfellow, Outre-Mer)
- “Music is what life sounds like.” (Eric Olson)

# Neurological links between music and language

- Music and particularly rhythm can help the recovery of patients who have lost the faculty of speech (aphasia) after severe head injuries. (Oliver Sacks, 2007)

- "Music is organized sound, but the organization has to involve some element of the unexpected or it is emotionally flat and robotic. Too much organization may technically still be music, but it would be music that no one wants to listen to. Scales, for example, are organized, but most parents get sick of hearing their children play them after five minutes." (Levitin, 2006, p.169)

**→ creativity is needed**

# There is a strong link between language and music

- Dimensions:
  - Melody - coherence
  - Rhythm
  - Pauses
  - Dissonances/divergences – consonances/convergences
  - Multivocality
  - Interanimation
  - Polyphony

# The Polyphonic Model of discourse

# Dialogism

- Basis for the CSCL paradigm (Koschman, 1999)
- "… Any true understanding is dialogic in nature" (Voloshinov-Bakhtin, 1973)
- Opposed to de Saussure ideas:
  - Real life dialog should be the focus, not written text
  - Words are not arbitrary
- Utterances (not sentences) should be the unit of analysis
- Speech genres
- Polyphony
- Inter-animation of voices

# Polyphony and counterpoint

- Concept derived from classical music
  - “These are different voices singing variously on a single theme. This is indeed 'multivoicedness,' exposing the diversity of life and the great complexity of human experience “ (Bakhtin, 1984)
- Multiple voices – each utterance contains multiple voices
- Voices inter-animate in an unmerged way:
  - “a plurality of independent and unmerged voices and consciousnesses” (Bakhtin)
- 'Everything in life is counterpoint, that is, opposition,' (Bakhtin, 1984)

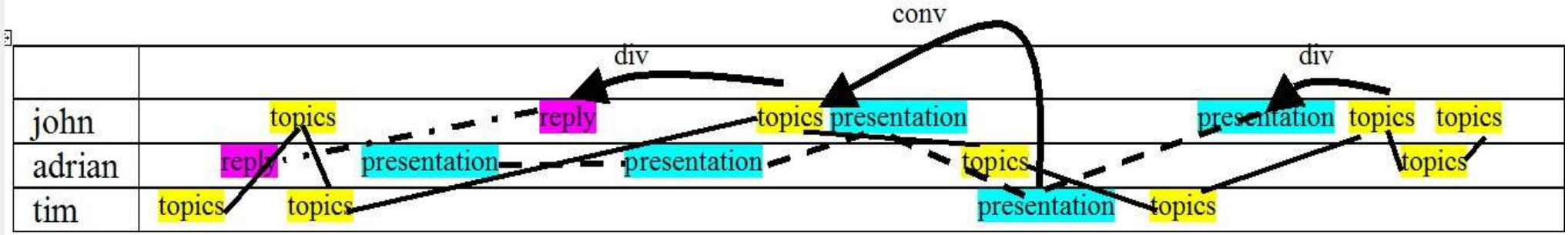
# Polyphony

- A group of participants that, each of them keeps their individuality, personality, creativity, but also collaborate to achieve a common goal, trying to solve dissonances
- A merge of:
  - Unity vs. Difference
  - Melody (longitudinal) and Harmony (transversal)
  - Dissonances – Consonances → Cycles – centrifugal/centripetal forces
- Inter-animation of voices – inter-animation patterns

# The Polyphonic Model

- Polyphony = Model of collaboration and interaction (Trausan-Matu, Stahl, and Zemel, 2005)
- Human communication in knowledge construction and collaboration are processes in which words and other utterances are linked in parallel threads which interact similarly to voices in polyphonic music
- Repetition, pauses, rhythm, and the "game" of dissonances/consonances are essential

| Nr | Ref | Time | User | Text |
|---|---|---|---|---|
| 17 | | 10.26.25 | tim | You discussed about a topic separation |
| 18 | 15 | 10.26.37 | adrian | First of all, the reply method is cumbersome |
| 19 | 17 | 10.26.50 | john | yes.. because we did not like the way the topics were presented in concert chat |
| 20 | 18 | 10.26.56 | john | yes !! |
| 21 | 20 | 10.27.04 | john | i hate double-clicking! |
| 22 | 20 | 10.27.18 | tim | and how can we find topics ? |
| 23 | 18 | 10.27.26 | adrian | What bothers me is the linear presentation of the discussin |
| 24 | 23 | 10.27.43 | john | Yep |
| 25 | 18 | 10.27.46 | adrian | and double-clicking too |
| 26 | | 10.27.54 | tim | You mean u want something like a chat forum ? :D |
| 27 | 24 | 10.27.58 | john | and the reply-to facility is supposed to help you |
| 28 | 18 | 10.28.15 | adrian | i'd like a tree presentation more |
| 29 | 18 | 10.28.38 | adrian | or maybe multiple chat columns, for each chat sub-thread |
| 30 | 27 | 10.28.58 | john | but it is really difficult to use in real-time because there are so many topics discussed which intertwine each other |
| 31 | 28 | 10.29.18 | john | i subscribe to a tree-like presentation form |
| 32 | P 30 | 10.29.20 | adrian | yes, that's why a clear separation of topics is needed |
| 33 | 31 | 10.29.47 | adrian | this is easy to implement, no problem here :) |
| 34 | 30 | 10.29.49 | tim | You need also a clever visual representation |
| 35 | 30 | 10.30.05 | tim | you'll need also a clever visual interface |
| 36 | | 10.30.22 | tim | Who decides the topics ? |
| 37 | 33 | 10.30.33 | john | i suppose you are refering to the visual representation, right ? |
| 38 | 37 | 10.30.45 | john | What i would like is a clever way to separate the topics :) |
| 39 | 38 | 10.30.59 | john | not just doing ot myself, manually |
| 40 | 37 | 10.31.00 | adrian | Yeah |
| 41 | 39 | 10.31.44 | adrian | When you start a new thread (a new message, non-related to other message), the app can assume a new topic |
| 42 | 39 | 10.31.46 | john | i would like the application to be able to detect w topic change all by itself |
| 43 | 42 | 10.32.01 | tim | That right |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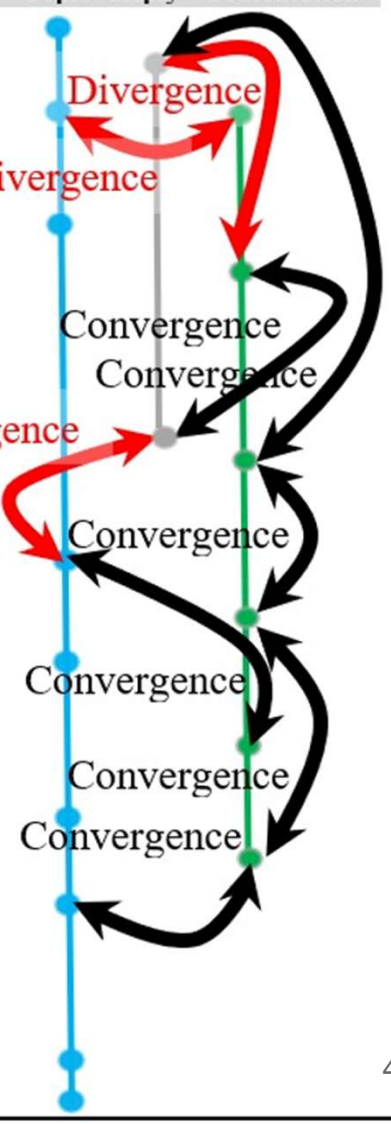


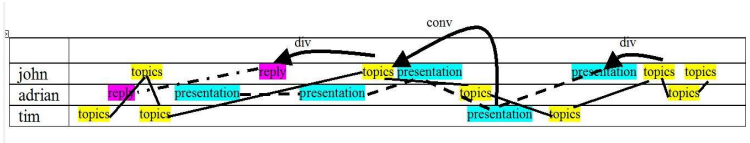
# Polyphonic inter-animation

# Polyphony- An Unitary Model of Human Communication

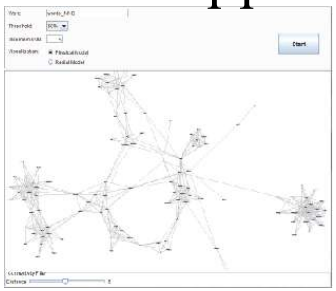
- Mediation:
  - Using natural language (words) in
    - texts
    - hypertexts
    - discussion forums
    - conversations

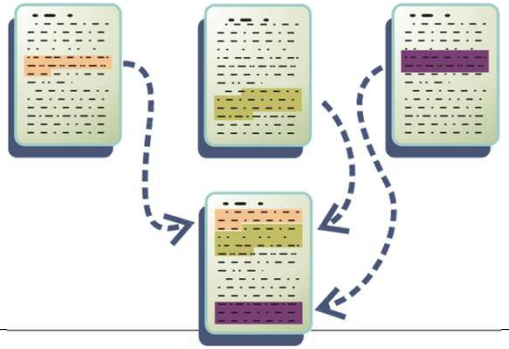
  but also
  - Non verbal communication (e.g. gestures)

# Polyphony- An Unitary Model of Group Communication




- Applicable to:
  - Small groups (e.g. virtual teams collaborating by chat or forums – the INTER-ANIMATION phenomenon appears)
  - Large groups – social networks
  - Global level - intertextuality





# Words, voices and threads

- Positions assigned to participants – voices in Bakhtin's sense
- Additional voices – frequent concepts – repeated words become voices, stronger or weaker (Trausan-Matu, Stahl, & Sarmiento, 2006)
- Voices continue and influence each other through explicit or implicit links.
- Voices correspond to chains or threads of utterances:
  - repeated words
  - lexical chains
  - co-references
  - reasoning or argumentation
  - rhetorical schemas

# Inter-animation patterns
## (Trausan-Matu, Stahl, & Sarmiento, 2006)

- Longitudinal
  - Adjacency pairs
  - Repetitions
  - Elaboration
  - Convergence
  - Cumulative talk
  - Repair
- Transversal, differential
  - Dissonance

# The Polyphonic Analysis Method

# The Polyphonic Analysis Method

- Operationalized qualitative method (Trausan-Matu et al. 2005, 2007) developed and used for several purposes:
  - investigating collaborative knowledge construction and learners' participation in collaborative chat sessions (Trausan-Matu et al. 2007b; Trausan-Matu 2013) - sessions performed at VMT and UPB, and also face-to-face collaboration in a Japanese class (Trausan-Matu 2013).
  - identification of the artifacts that enable problem solving (Trausan-Matu 2012)
  - identification of important (pivotal) moments in conversations (Trausan-Matu 2013).

# Steps of the Polyphonic Analysis Method

**1.** **utterances** are delimited

**2.** **links** are identified between utterances starting from:

- **Repetitions of words and phrases,**
- **Adjacency pairs,**
- **Justification links,**
- **Co-references, etc.**

As result, a ***graph of utterances*** may be constructed,

**3.** Identification of **voices** - **threads** (re-voicings (echoes), should be detected starting from repeated important words

**4.** identification of **inter-animation patterns** among voices, starting from discourse markers („but", „nevertheless", „therefore", „also", „moreover", etc.):

- **Divergences**
- **Convergeces**

# The Polyphonic Analysis applications

- Chat conversations with multiple participants for:
  - CSCL
  - Fostering creativity – **brainstorming,** synectics
- F2F collaborative learning
- Analysis of rhythm and pauses
- Metacognition (conversation & essays)
- Intertextuality
- Sonification

# Computer Support for the Polyphonic Analysis

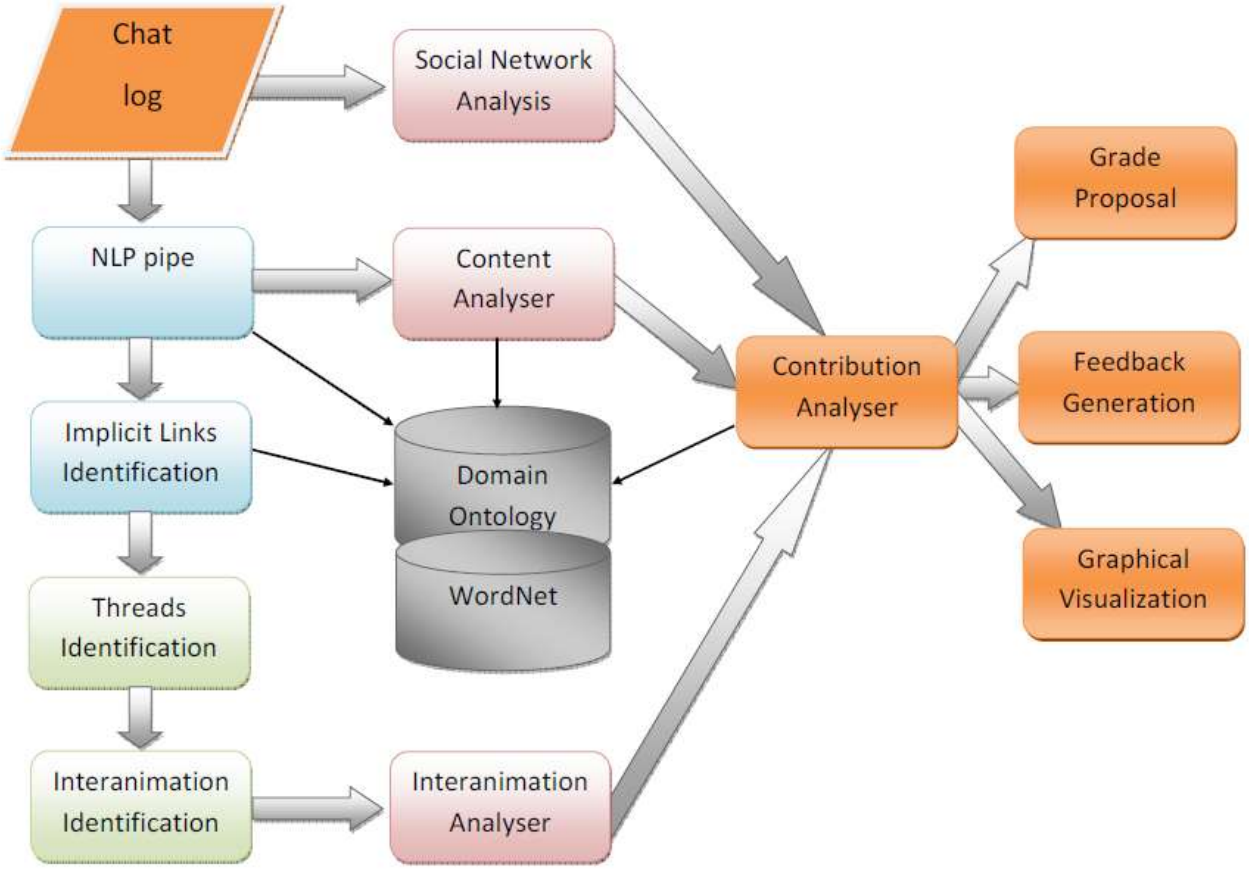
# Systems based on the polyphonic model

- The “Polyphony” system (Trausan-Matu, Rebedea, and all, 2007)
- ASAP (Dascalu, Chioasca and Trausan-Matu, 2008)
- PolyCAFe (Trausan-Matu and Rebedea, 2010; Trausan-Matu, Dascalu, and Rebedea, 2015)
- Collaboration regions detection (Banica, Trausan-Matu and Rebedea, 2011)
- Detection of the Important moments (Chiru and Trausan-Matu, 2012)
- ReaderBench (Dascalu, Trausan-Matu, Dessus, 2012)
- Sonification (Trausan-Matu and Calinescu, 2013)

# LTfLL - EU FP7 Project (2008-2011) and NSF Virtual Math Teams Project

http://mathforum.org
http://ltfll-lin.code.ro:11080/concertChat

- Language Technologies for Lifelong Learning
  - Netherlands, France, United Kingdom, Germany, Austria, Romania, Bulgaria
- PolyCAFe system (Polyphony-based Collaboration Analysis and Feedback generation)
  - The system has been validated with students and tutors in
    - University of Manchester, UK
    - Politehnica University of Bucharest, Romania


Chat log
Social Network Analysis
NLP pipe
Content Analyser
Implicit Links Identification
Domain Ontology
WordNet
Threads Identification
Interanimation Identification
Interanimation Analyser
Contribution Analyser
Grade Proposal
Feedback Generation
Graphical Visualization